VICTORY 2027

LIGHT

LECTURE-04 PHYSICS

Shailendra sir

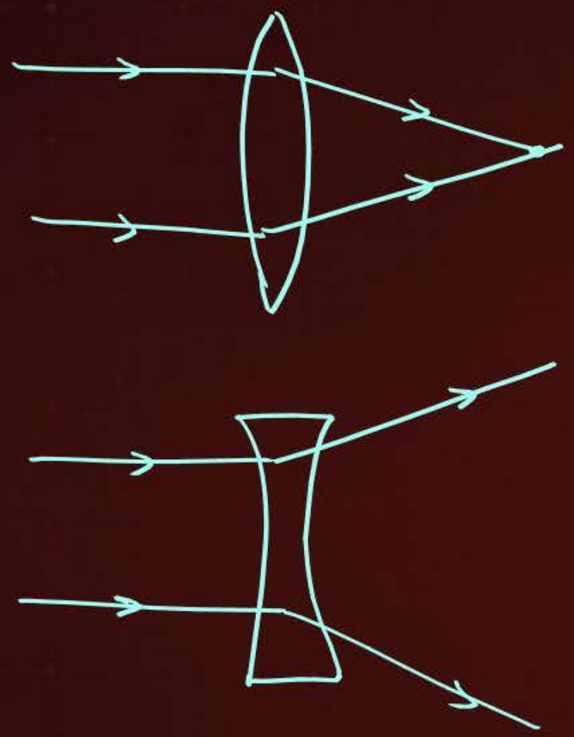


Topics To Be Covered

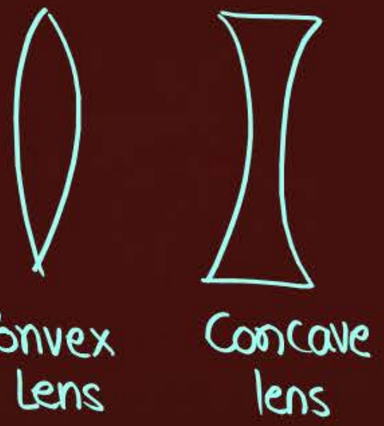
1. REVISION
2. LENS
3. FORMATION OF DIFFERENT TYPES OF IMAGES BY LENS
4. QUESTIONS


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REVISION







LENS

A lens is a transparent refracting medium bounded by either the two spherical surfaces or one surface spherical and other surface plane.

A plane surface can be treated as a spherical surface of infinite radius of curvature.

Kind of lenses:

(1) converging or convex lens

(2) diverging or concave lens

CONVERGING OR CONVEX LENS

A convex lens is thick in its middle and thin at the periphery. In other words, a lens which bulges out in the middle, is the convex lens. A light beam converges on passing through such a lens, so it is also called the converging lens.



DIVERGING OR CONCAVE LENS

A concave lens is thick at its periphery and thin in the middle. In other words, a lens which is bent inwards in the middle, is the concave lens. Such a lens diverges the light rays incident on it, so it is also called the diverging lens.



IMPORTANT POINT

- Both the concavo-convex and the convexo-concave lenses have one surface convex and the other surface concave, but they differ in their shape and action.
- A concavo-convex lens is thicker in the middle and has a converging action on a light beam.
- While a convexo-concave lens is thinner in the middle and has a diverging action on a light beam.



ACTION OF A LENS AS A SET OF PRISMS

- The lens can be considered as being made up of a rectangular slab at the centre and one prism on either side of it.



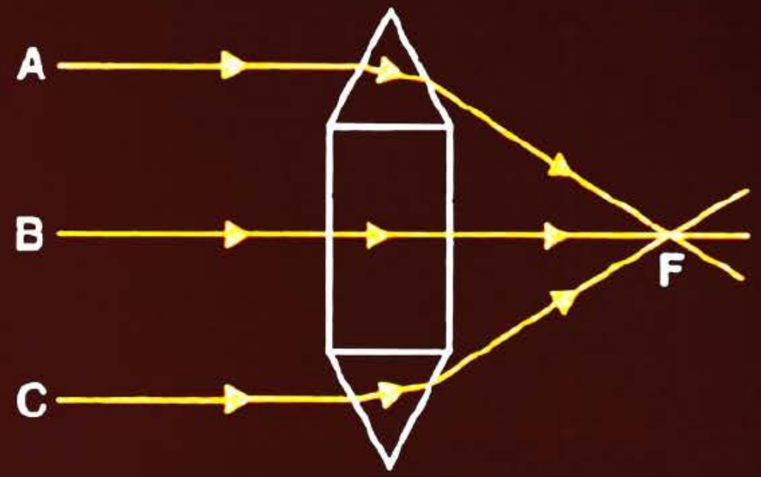
(a) Convex lens



(b) Concave lens

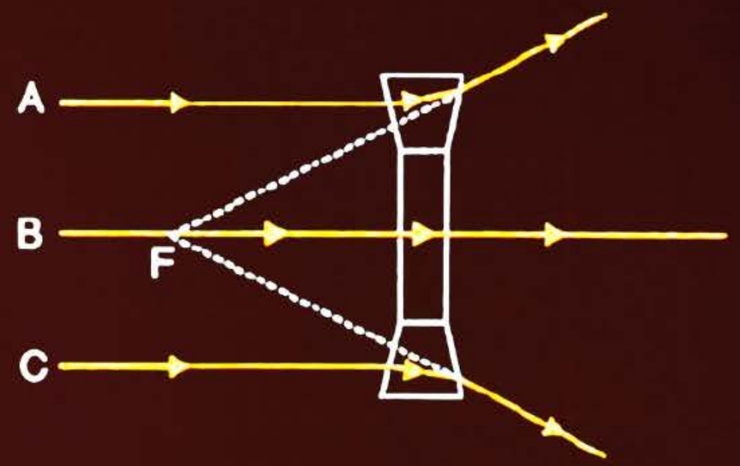
CONVERGENT ACTION OF A CONVEX LENS

- A ray of light incident on a prism, on refraction through it, always bends towards the base of prism.
- Therefore the prism in the upper part of convex lens bends the incident ray A downwards, while the prism in the lower part of convex lens bends the incident ray C upwards.
- The central part which is a parallel sided glass slab passes the ray B normally incident on it, undeviated.
- Thus the set of prisms forming a convex lens converges the parallel rays to a point F . Therefore a convex lens has a converging action on the incident light rays.



DIVERGENT ACTION OF A CONCAVE LENS

- The prism in the upper part of the concave lens bends the incident ray A upwards i.e., towards its base, while the prism in the lower part of the concave lens bends the incident ray C downwards i.e., towards its base.
- The central part, which is a parallel sided glass slab, passes the normally incident ray B undeviated.
- Thus, the set of prisms forming a concave lens diverges the parallel rays as if they are coming from a common point F situated on the side of rays incident on the lens.
- Therefore, a concave lens has a diverging action on the incident light rays.

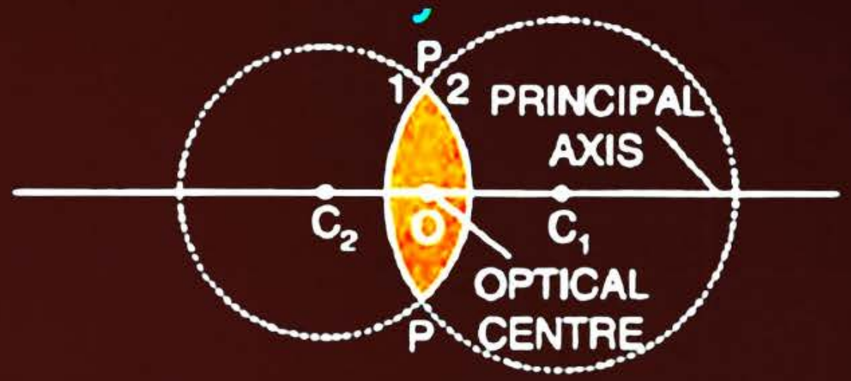


TECHNICAL TERMS RELATED TO A LENS

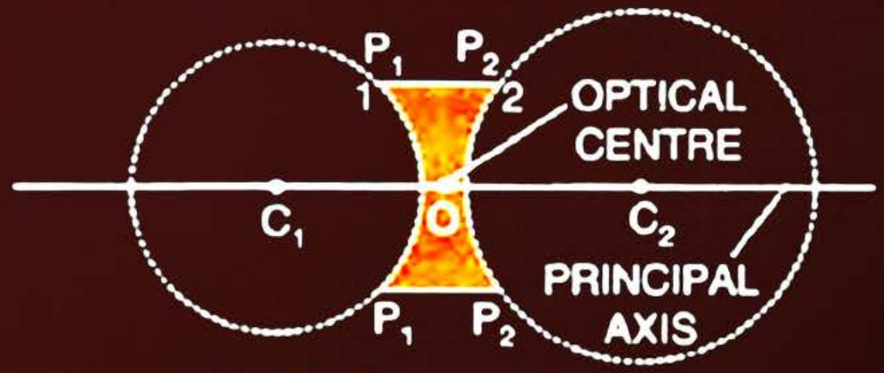
CENTRE OF CURVATURE:

A lens has two surfaces. Each surface of the lens is a part of a sphere. The centre of the sphere whose part is the lens surface, is called the centre of curvature of that surface of the lens.

Since a lens has two spherical surfaces, so there are two centres of curvature of a lens.



(a) Convex lens

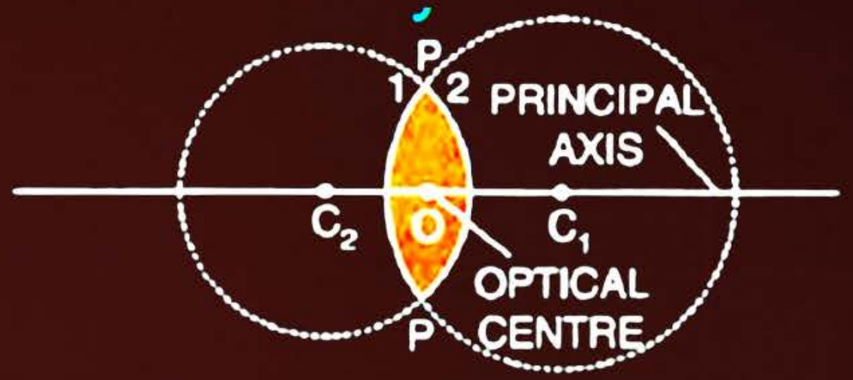


(b) Concave lens

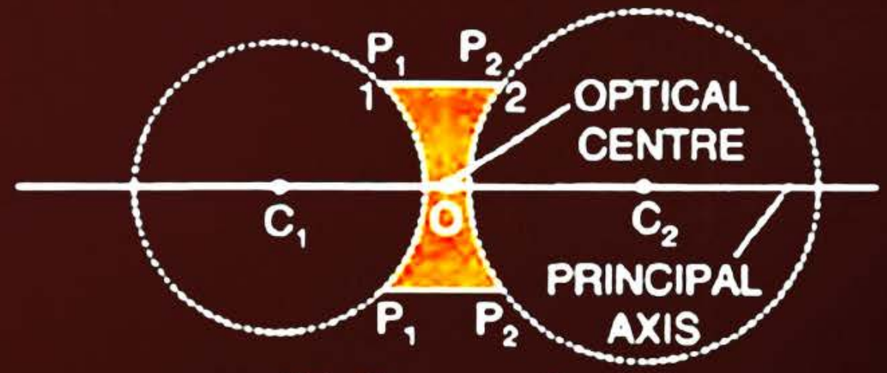
TECHNICAL TERMS RELATED TO A LENS

RADIUS OF CURVATURE:

The radius of the sphere whose part is the lens surface, is called the radius of curvature of that surface of the lens.



(a) Convex lens

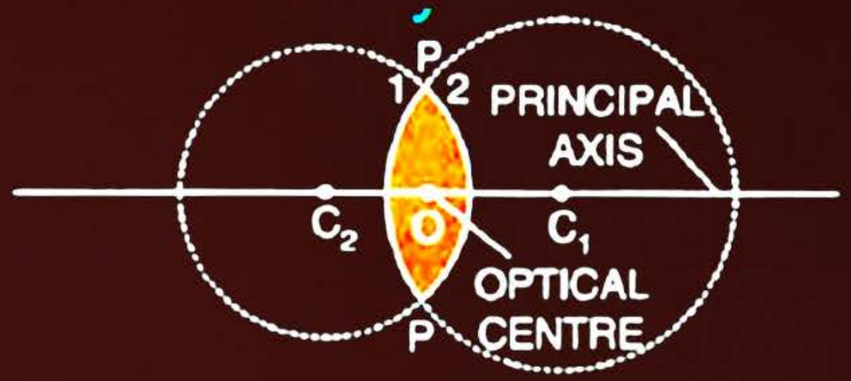


(b) Concave lens

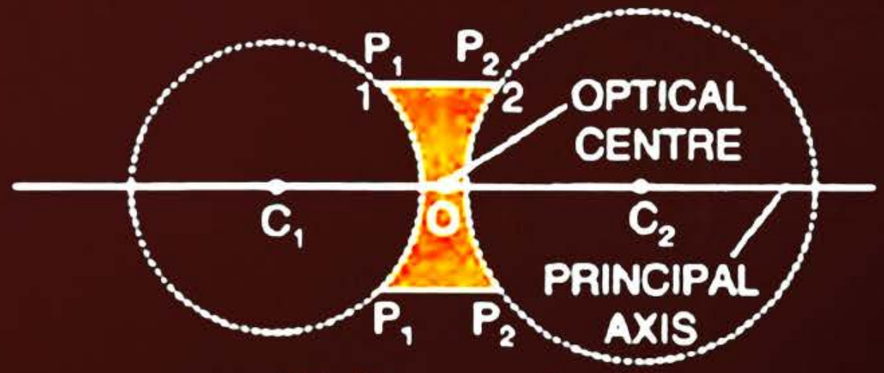
TECHNICAL TERMS RELATED TO A LENS

PRINCIPAL AXIS:

It is the line joining the centres of curvature of the two surfaces of the lens. It can extend on either side of the lens.



(a) Convex lens

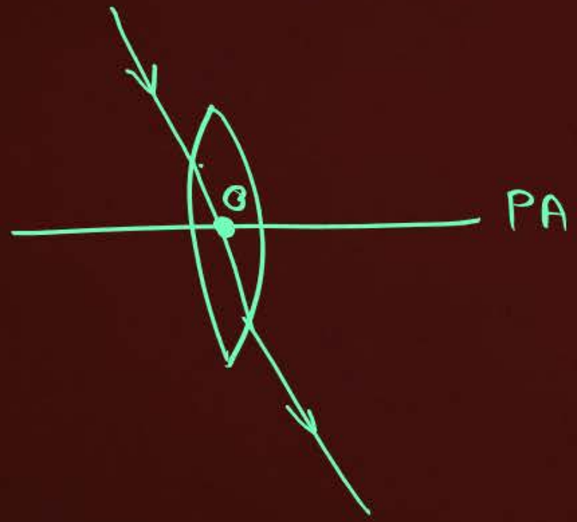


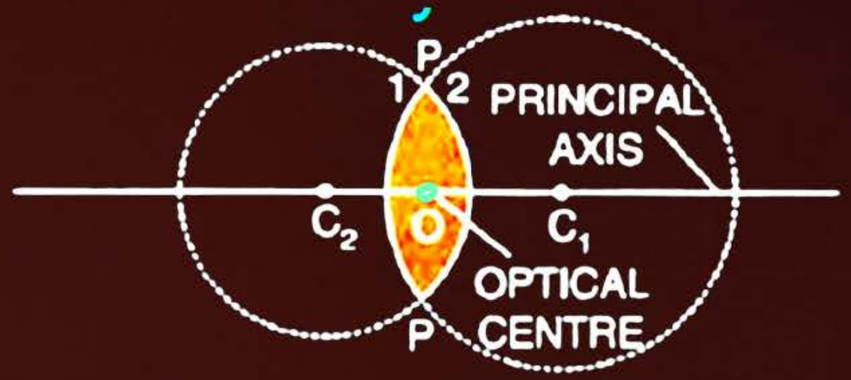
(b) Concave lens

TECHNICAL TERMS RELATED TO A LENS

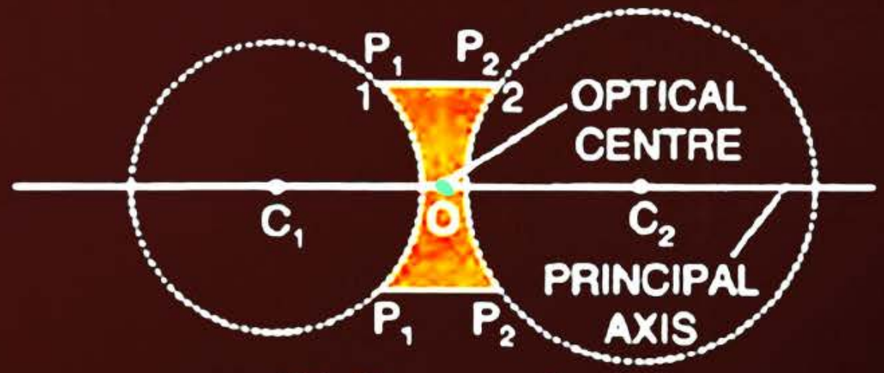
OPTICAL CENTRE:

It is a point on the principal axis of the lens such that a ray of light passing through this point emerges parallel to its direction of incidence. It is marked by the letter O in Fig. The optical centre is thus the centre of lens.





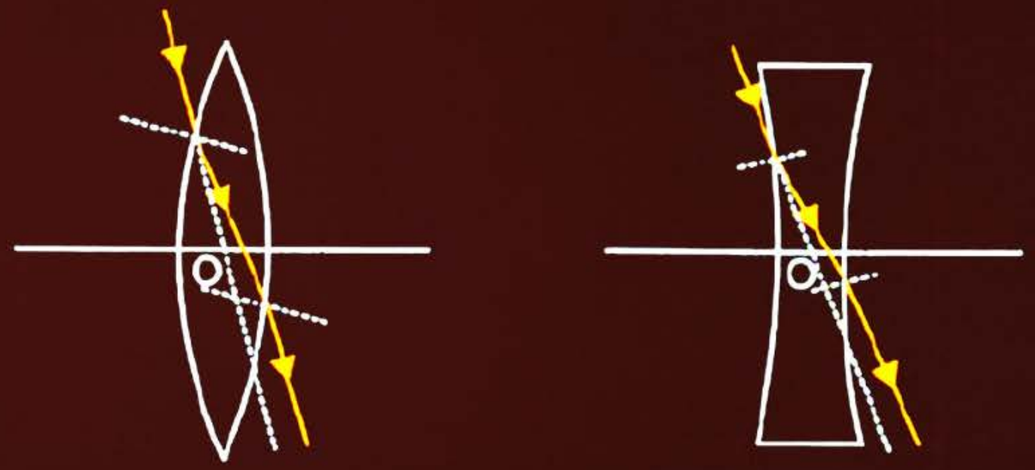
(a) Convex lens



(b) Concave lens

TECHNICAL TERMS RELATED TO A LENS

Since the central portion of a thick lens can be considered to be a parallel sided glass slab, therefore a ray of light incident at the central portion of the lens, while passing through the optical centre O , is slightly displaced parallel to its original direction. In Fig, the emergent ray is thus parallel to the incident ray.

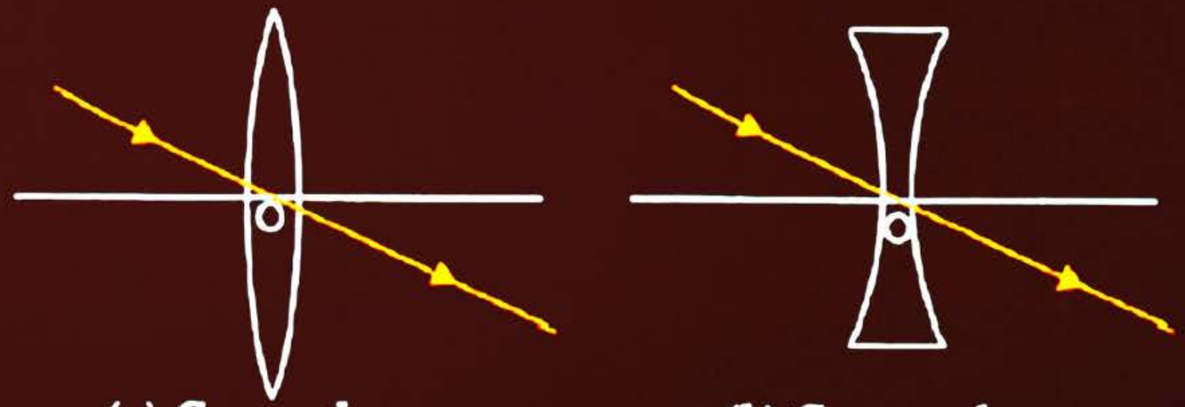


(a) Convex lens *(b) Concave lens*

TECHNICAL TERMS RELATED TO A LENS

Generally the lens is thin, so the lateral shift is small enough and it can be ignored. Therefore, a ray of light directed towards the optical centre of a thin lens can be considered to pass undeviated and undisplaced.

Optical centre of a thin lens is the point on the principal axis of lens such that a ray of light directed towards it, passes undeviated through it.

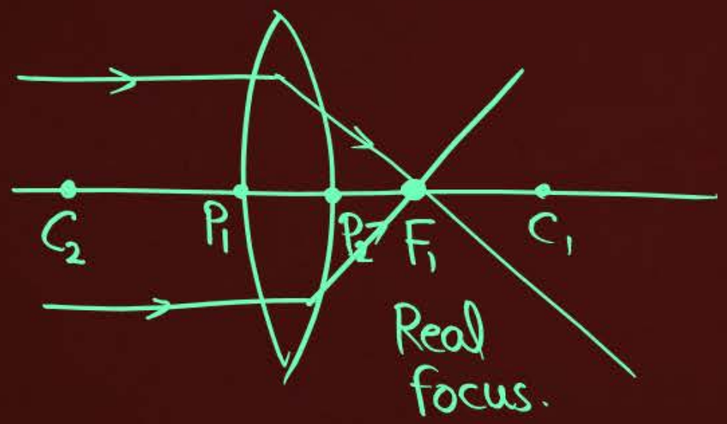


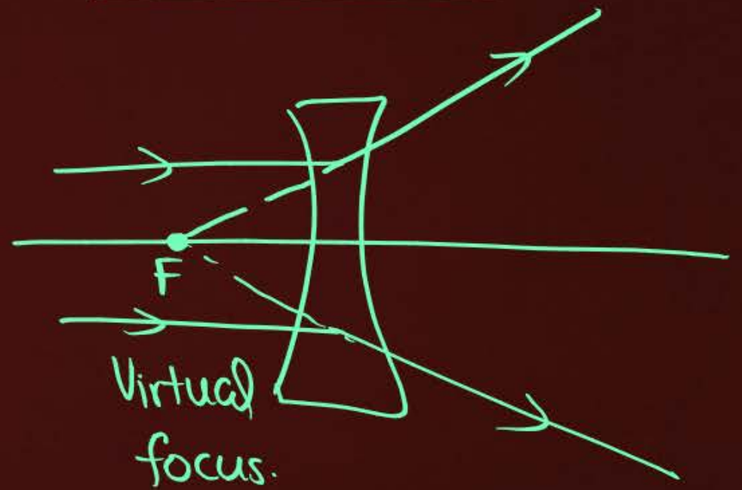
(a) Convex lens *(b) Concave lens*

TECHNICAL TERMS RELATED TO A LENS

PRINCIPAL FOCI:

A light ray can enter a lens from either side, therefore, a lens has two principal foci. If medium is same on either side of the lens, the two foci are situated at equal distances from the optical centre, one on either side of the lens. These are known as the first focal point (or first focus) F_1 and the second focal point (or second focus) F_2.



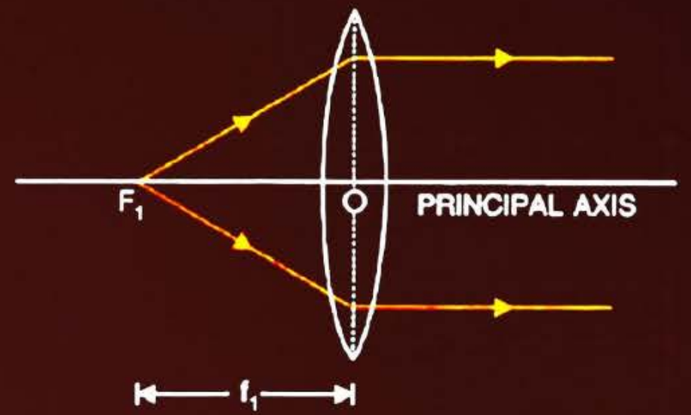


TECHNICAL TERMS RELATED TO A LENS

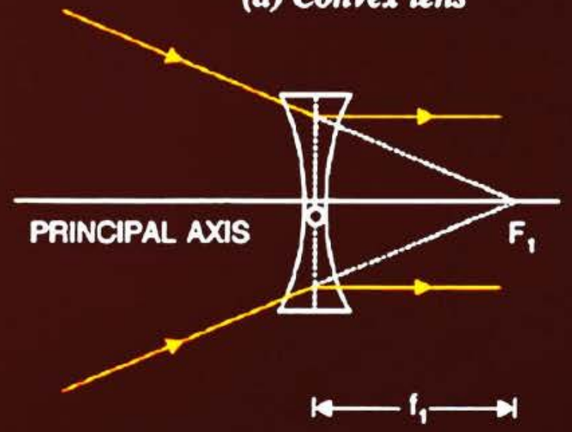
FIRST FOCAL POINT:

For a convex lens, the first focal point is a point F_1 on the principal axis of the lens such that the rays of light coming from it, after refraction through the lens, become parallel to the principal axis of the lens.

For a concave lens, first focal point is a point F_1 on the principal axis of the lens such that the incident rays of light appearing to meet at it, after refraction from the lens become parallel to the principal axis of the lens.



(a) Convex lens



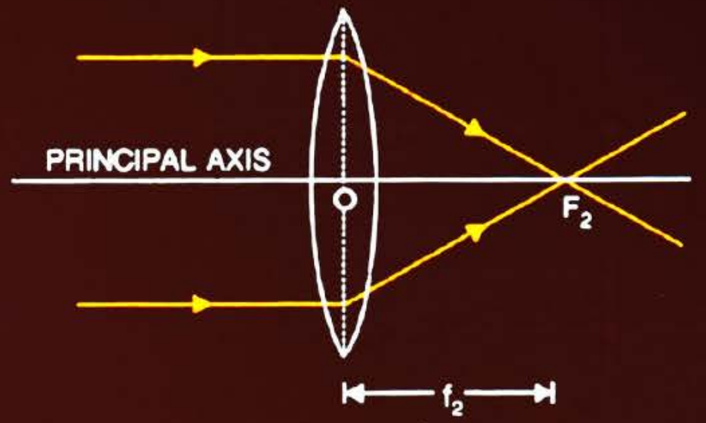
(b) Concave lens

TECHNICAL TERMS RELATED TO A LENS

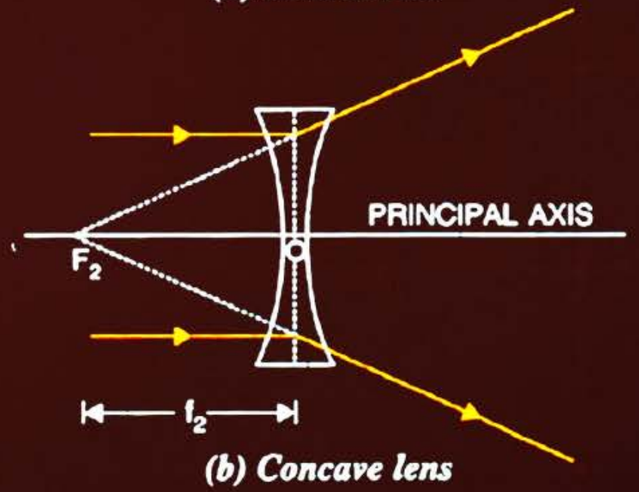
SECOND FOCAL POINT:

For a convex lens, the second focal point is a point F_2 on the principal axis of the lens such that the rays of light incident parallel to the principal axis, after refraction from the lens, pass through it.

For a concave lens, second focal point is a point F_2 on the principal axis of the lens such that the rays of light incident parallel to the principal axis, after refraction from the lens, appear to be diverging from this point.



(a) Convex lens



(b) Concave lens

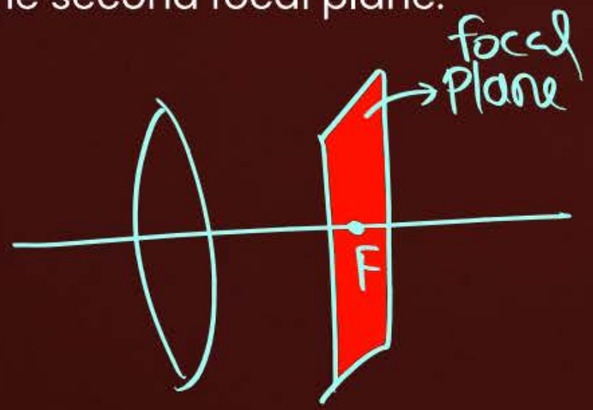
TECHNICAL TERMS RELATED TO A LENS

FOCAL PLANE:

A plane normal to the principal axis, passing through the focus, is called the focal plane. A lens has two focal planes.

(i) First focal plane: A plane passing through the first focal point and normal to the principal axis of the lens, is called the first focal plane.

(ii) Second focal plane: A plane passing through the second focal point and normal to the principal axis of the lens, is called the second focal plane.



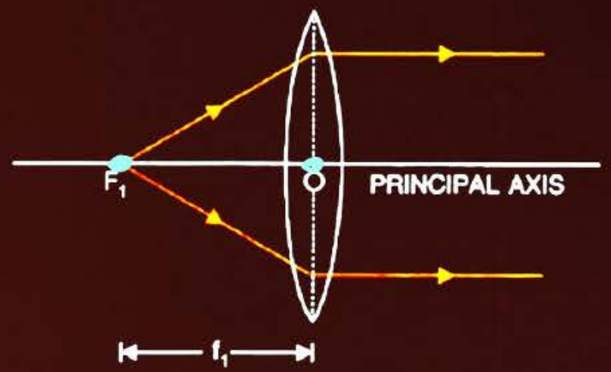
TECHNICAL TERMS RELATED TO A LENS

FOCAL LENGTH:

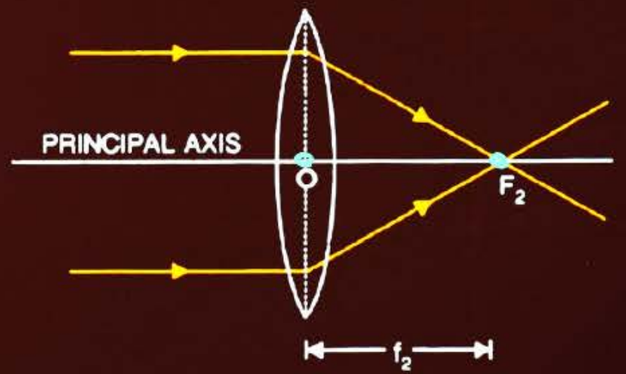
The distance of focus (or focal point) from the optical centre of lens, is called its focal length. A lens has two focal lengths.

(i) First focal length: The distance from the optical centre O of the lens to its first focal point F_1 is called the first focal length f_1 of the lens. In Fig, it is shown as $OF_1 = f_1$.

(ii) Second focal length: The distance from the optical centre O of the lens to the second focal point F_2 is called the second focal length f_2 of the lens. In Fig, it is shown as $OF_2 = f_2$.



(a) Convex lens



(a) Convex lens

SOME IMPORTANT POINTS

(1) If the medium on both sides of a lens is same, its first and second focal lengths are equal, i.e., $f_1 = f_2$ (numerically).

(2) Usually, when we say focus, we mean the second focal point. Hence the focal length of a lens implies the second focal length of the lens.

(3) A convex lens has a real focus (because the parallel rays incident on a convex lens actually pass through this point), while in a concave lens the focus is virtual (because the parallel rays incident on a concave lens do not actually pass through this point, but they appear to diverge from this point).

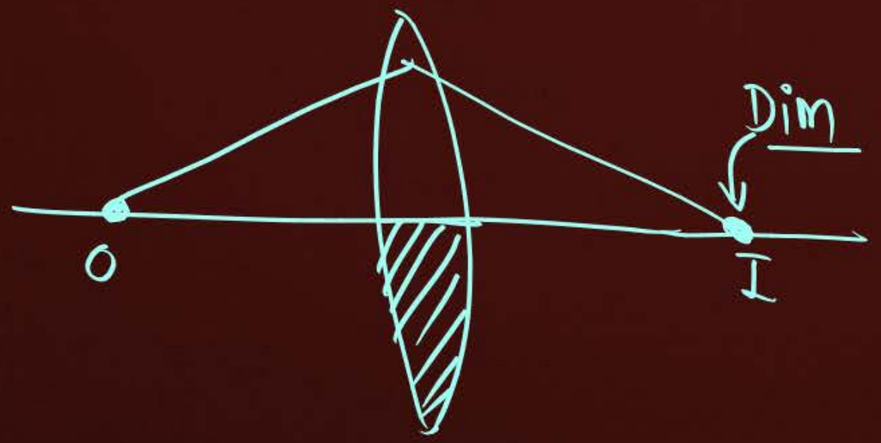
SOME IMPORTANT POINTS

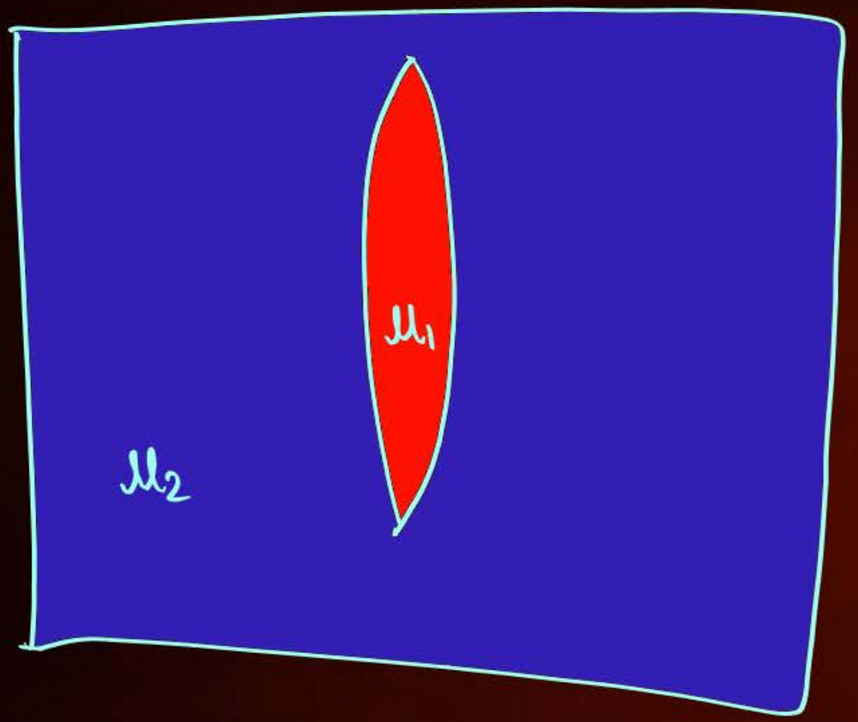
(4) The focal length of a lens depends on the following two factors:

(i) The refractive index of the material of lens relative to its surrounding medium. If a lens is placed in water instead of air, its focal length increases.

(ii) The radii of curvature of the two surfaces of lens. A thick lens has less focal length than a thin lens of same material.

(5) If a part of the lens is covered, its focal length remains unchanged, only the amount of light entering the lens decreases due to which the intensity of image decreases but the position, size and nature of image formed by it do not change.





Lens Maker's Formula.

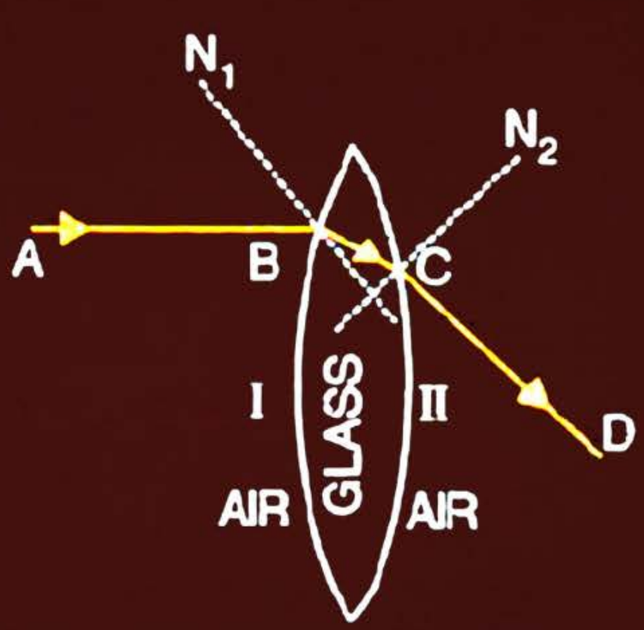
$$\frac{1}{f} = \left(\frac{\mu_1}{\mu_2} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

DIFFERENCE BETWEEN CONVEX AND CONCAVE LENS

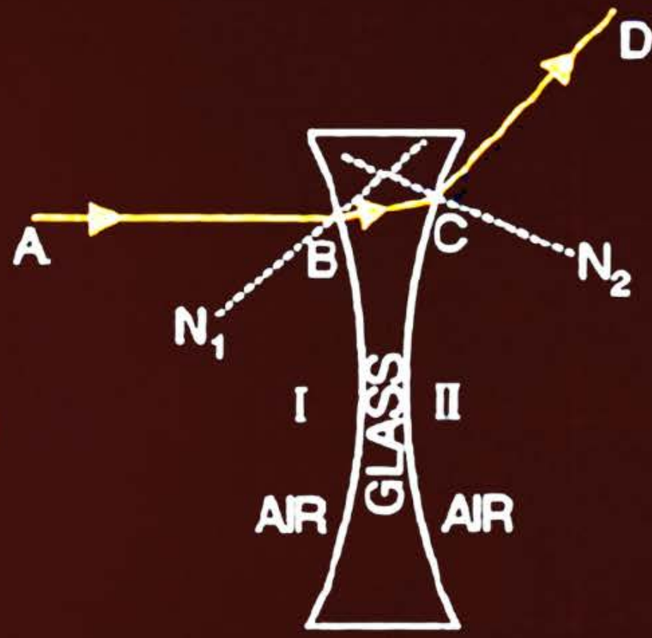
HW

	Convex Lens		Concave Lens
1.	It is thick in the middle and thin at its periphery.	1.	It is thin in the middle and thick at its periphery.
2.	It converges the incident rays towards the principal axis.	2.	It diverges the incident rays away from the principal axis.
3.	It has a real focus.	3.	It has a virtual focus.

REFRACTION OF LIGHT THROUGH THE EQUI-CONVEX AND EQUI-CONCAVE LENSES



(a) Convex lens



(b) Concave lens

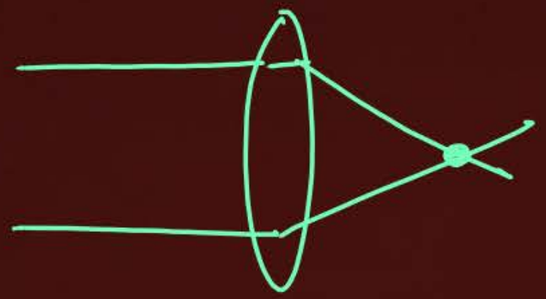
QUESTIONS

A ray of light after refraction through a lens emerges parallel to the principal axis of the lens. The incident ray passes through:

1. its optical centre
2. its first focus
3. its second focus
4. the centre of curvature of the first surface.

QUESTIONS

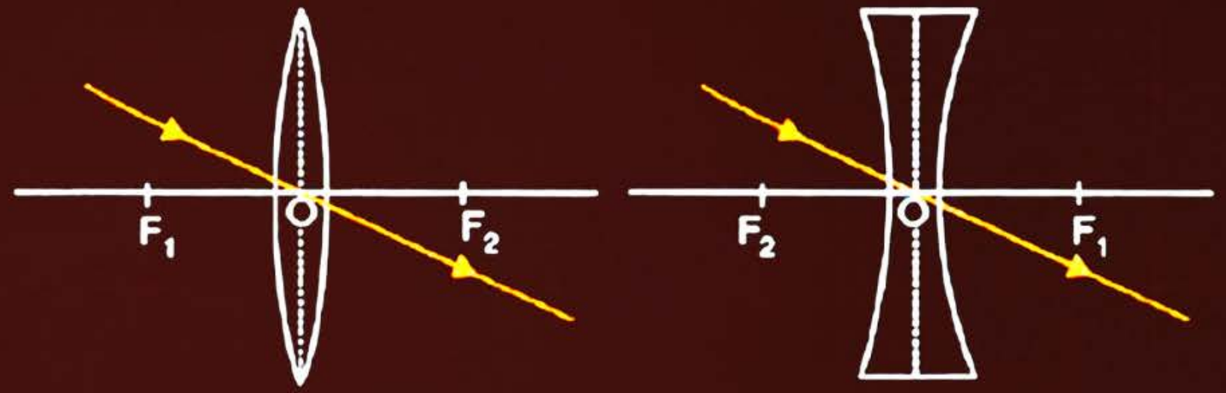
A ray of light incident on a lens parallel to its principal axis, after refraction passes through or appears to come from:



1. its first focus
2. its optical centre
3. its second focus
4. the centre of curvation of its second surface.

PRINCIPLES (OR CONSTRUCTION) RAYS FOR RAY DIAGRAMS

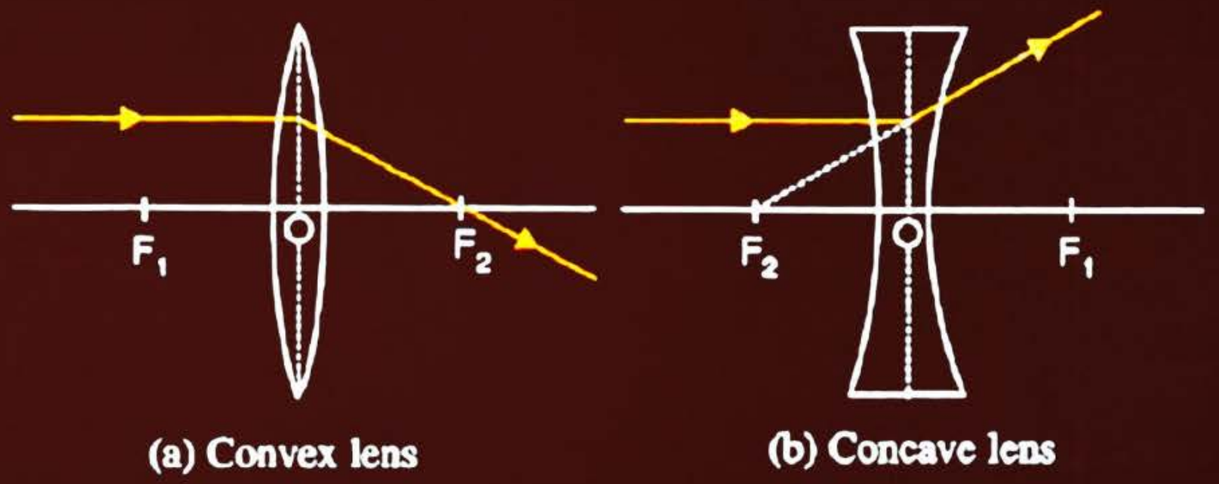
(1) A ray of light incident at the optical centre O of the lens passes undeviated through the lens respectively for the convex and concave lens.



(a) Convex lens (b) Concave lens

PRINCIPLES (OR CONSTRUCTION) RAYS FOR RAY DIAGRAMS

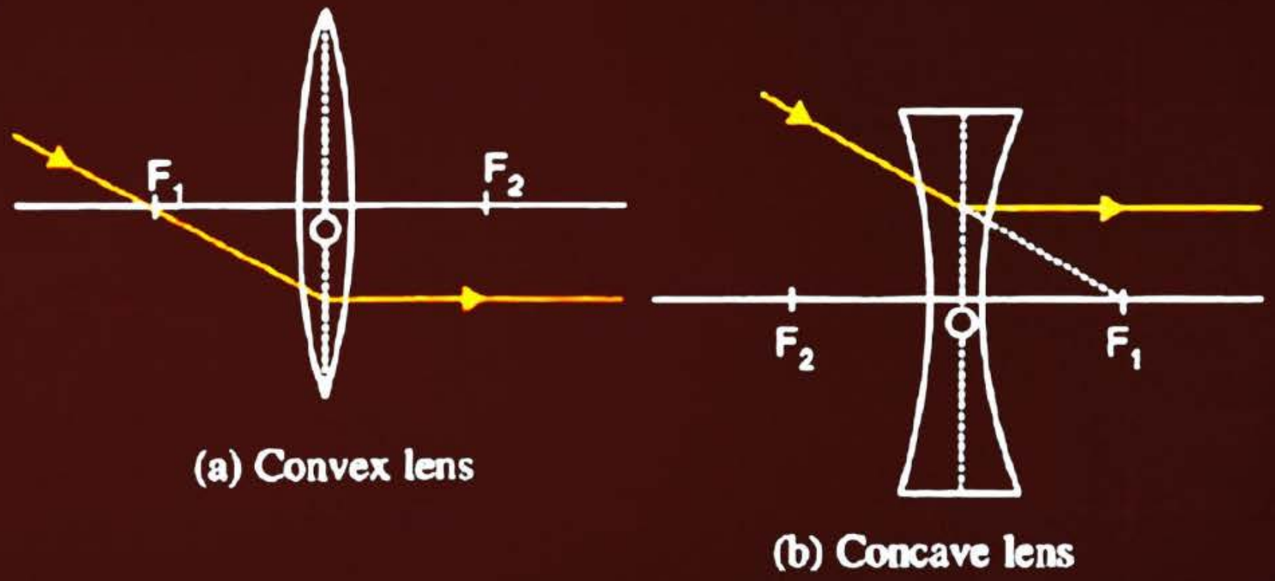
(2) A ray of light incident parallel to the principal axis of the lens, after refraction passes through the second focus F_2 (in a convex lens) or appears to come from the second focus F_2 (in a concave lens).



(a) Convex lens (b) Concave lens

PRINCIPLES (OR CONSTRUCTION) RAYS FOR RAY DIAGRAMS

(3) A ray of light passing through the first focus F_1 (in a convex lens) or directed towards the first focus F_1 (in a concave lens), emerges parallel to the principal axis after refraction



(a) Convex lens

(b) Concave lens

IMAGE

(1) **Real image:** If the rays from a point of object after refraction through the lens actually meet at a point, the image is real. If a screen is placed at this point, the image is obtained on it (i.e., a real image can be obtained on a screen).

(2) **Virtual image:** If the rays from a point of object after refraction through the lens do not actually meet at a point, but they appear to diverge from a point, the image is virtual. A screen placed at this point will not show any image on it (i.e., a virtual image cannot be obtained on a screen).

- However, the eye kept between the diverging rays is able to see this image because the eye lens being convex, converges the diverging rays to form the image on the retina of eye.

IMAGE

HW

Real → Inverted
Virtual → Erect

	Real Image		Virtual Image
1.	A real image is formed due to actual intersection of the rays refracted by the lens.	1.	A virtual image is formed when the rays refracted by the lens appear to meet if they are produced backwards.
2.	A real image can be obtained on a screen.	2.	A virtual image cannot be obtained on a screen.
3.	A real image is inverted with respect to the object. Example: The image of a distant object formed by a convex lens.	3.	A virtual image is erect with respect to the object. Example: The image of an object formed by a concave lens.

FORMATION OF DIFFERENT TYPES OF IMAGES BY A CONVEX LENS

Object between C and F

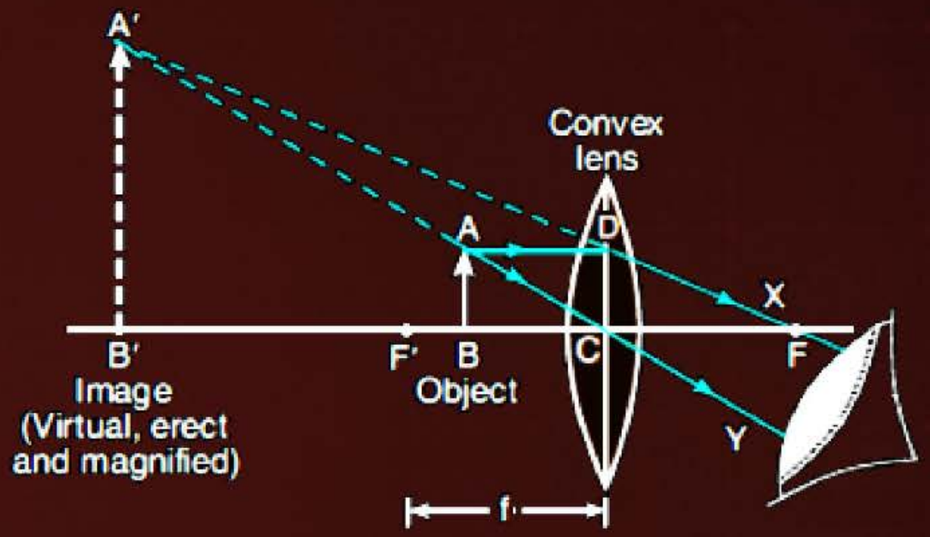
The image formed is:

(i) behind the object (on the left side of lens),

(ii) virtual and erect, and

(iii) larger than the object (enlarged or magnified).

Application: In this manner, a convex lens is used as a reading lens (i.e., a magnifying glass or a simple microscope) to form a magnified virtual image of a tiny object (such as a small letter of a book or a small part of a watch etc.).



FORMATION OF DIFFERENT TYPES OF IMAGES BY A CONVEX LENS

Object at F

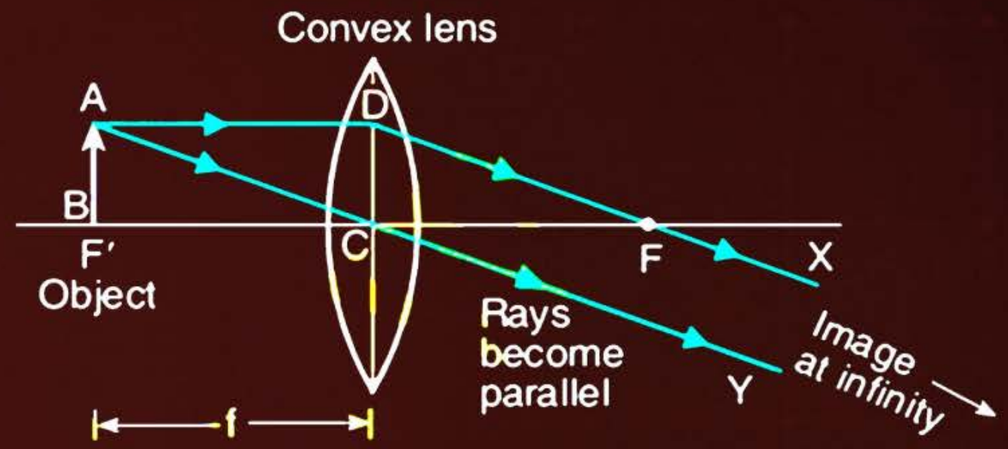
when an object is placed at the focus of a convex lens, the image formed is:

(i) at infinity,

(ii) real and inverted, and

(iii) highly enlarged.

Application: In this manner, a convex lens is used in the collimator of a spectrometer to obtain a parallel beam of light by placing the source of light at the focus of convex lens.



FORMATION OF DIFFERENT TYPES OF IMAGES BY A CONVEX LENS

Object between *f* and 2*f*

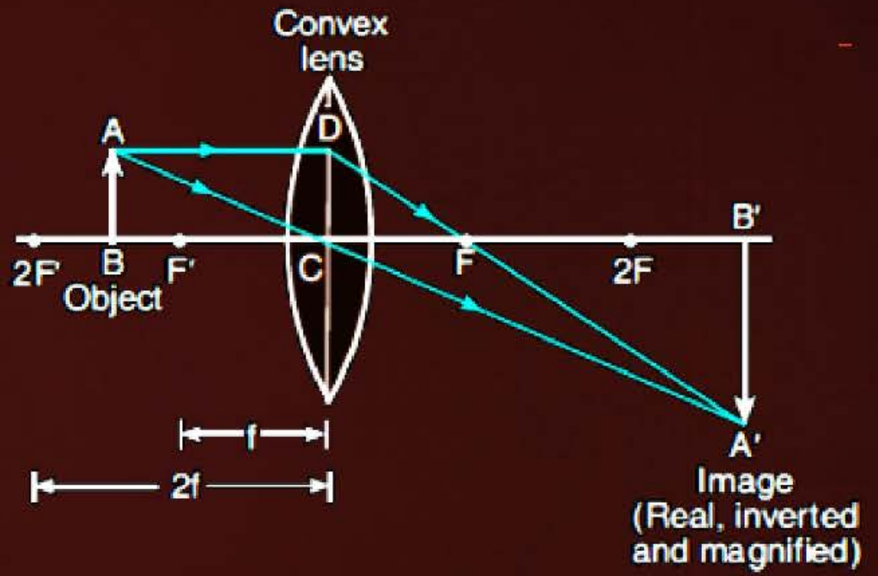
The image formed is:

(i) beyond 2F,

(ii) real and inverted, and

(iii) larger than the object (or magnified).

Application: In this manner, a convex lens is used in cinema and slide projectors. Here the magnified image is obtained on a screen placed at a large distance on the other side of the lens. Care is taken to put the slide (or film i.e., illuminated object) in front of the lens just beyond its focus in inverted position so as to obtain an erect and magnified image on the screen.



FORMATION OF DIFFERENT TYPES OF IMAGES BY A CONVEX LENS

Object at 2*f*

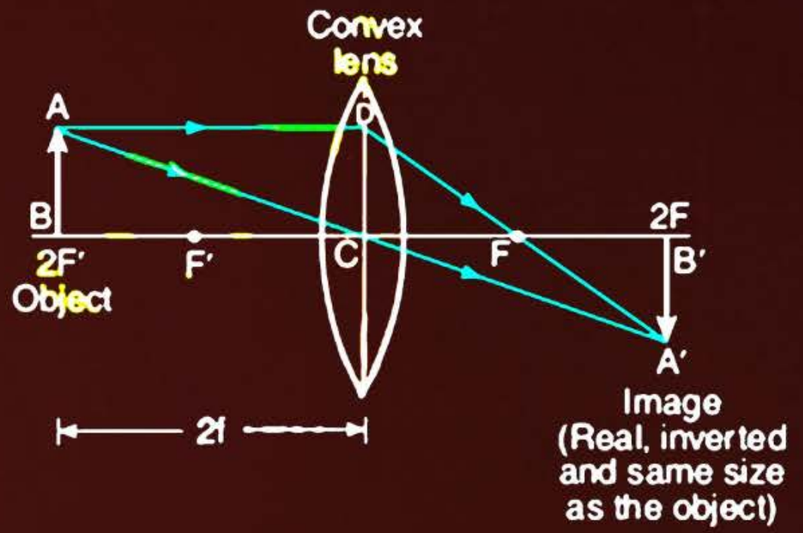
The image formed is:

(i) at a distance 2 f on the other side of the lens,

(ii) real and inverted, and

(iii) of the same size as the object.

Application: In this manner, a convex lens is used in a terrestrial telescope for erecting the inverted image formed by the objective lens.



FORMATION OF DIFFERENT TYPES OF IMAGES BY A CONVEX LENS

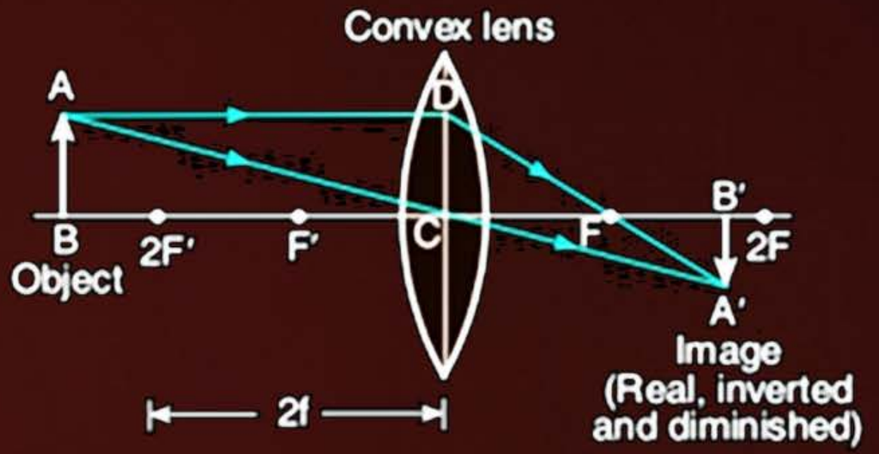
Object beyond 2*f*

The image formed is:

(i) between f and 2 f on the other side of the lens,

(ii) real and inverted, and

Application: A convex lens is used in this manner as a camera lens, when the object, not very far, is to be photographed.



FORMATION OF DIFFERENT TYPES OF IMAGES BY A CONVEX LENS

When the object is at infinity

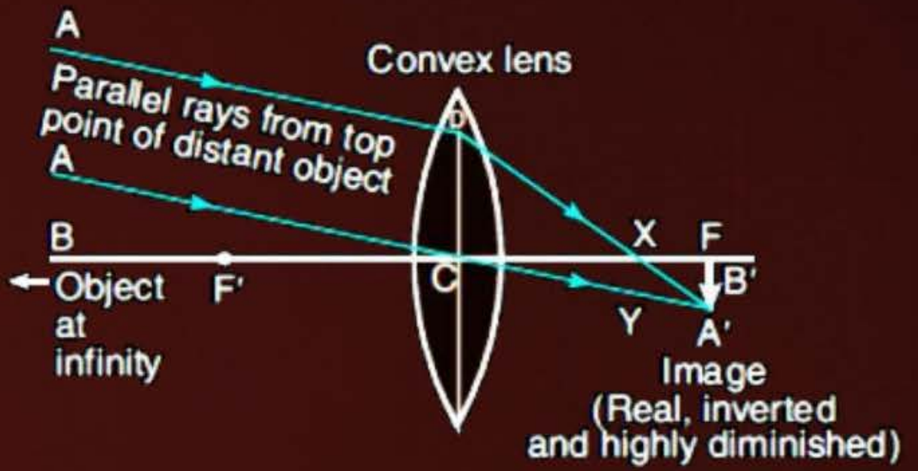
The image formed is:

(i) at the focus,

(ii) much sreal and inverted, and

(iii) maller than the object (or highly diminished

Application: In this manner, a convex lens is used either as a burning glass or a camera lens. To use the convex lens as a burning glass, the rays of light from sun (at infinity) are brought to focus on a piece of paper kept in the second focal plane of the lens. Due to sufficient heat of sun rays, the paper burns. In a camera, the object lies very far from the lens and the image is formed at the camera film which is at the second focus (or lies in the second focal plane) of the lens.



RELATIVE POSITIONS OF THE OBJECT AND IMAGE IN A CONVEX LENS

	Position of Object	Position of Image	Size of Image	Nature of Image	Application
1.	At infinity	At F_2	Highly diminished	Real and inverted	Burning glass
2.	Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	Real and inverted	Camera lens
3.	At $2F_1$	At $2F_2$	Same size	Real and inverted	Terrestrial telescope
4.	Between F_1 and $2F_1$	Beyond $2F_2$	Magnified	Real and inverted	Slide projector
5.	At F_1	At infinity	Highly magnified	Real and inverted	Collimator of spectrometer
6.	Between the lens and F_1	On same side, behind the object	Magnified	Virtual and upright	Magnifying glass

FORMATION OF IMAGE BY A CONCAVE LENS

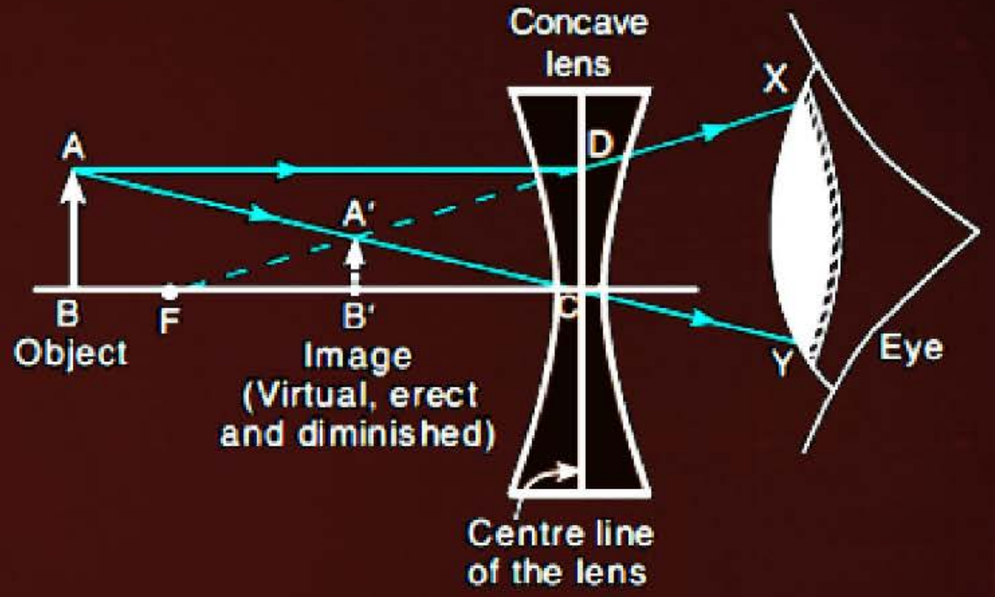
When an object is placed anywhere between optical centre (C) and infinity in front of a concave lens, the image formed is:

(i) between optical centre (C) and focus (F),

(ii) virtual and erect, and

(iii) diminished (smaller than the object).

Application: A concave lens is used in spectacles for the short-sighted persons in this manner.



FORMATION OF IMAGE BY A CONCAVE LENS

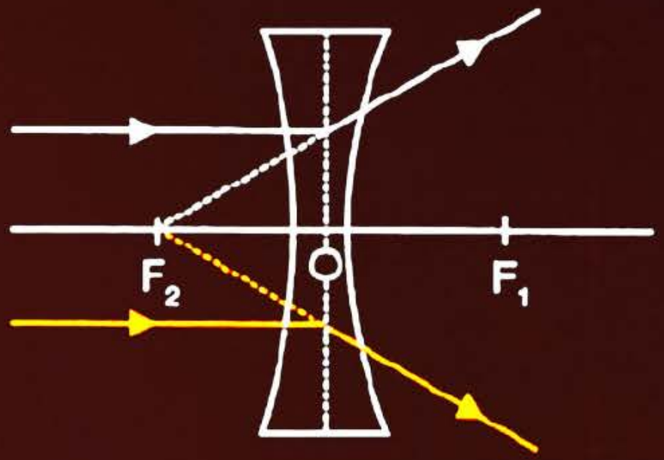
When an object is at infinity from a concave lens, the image formed is:

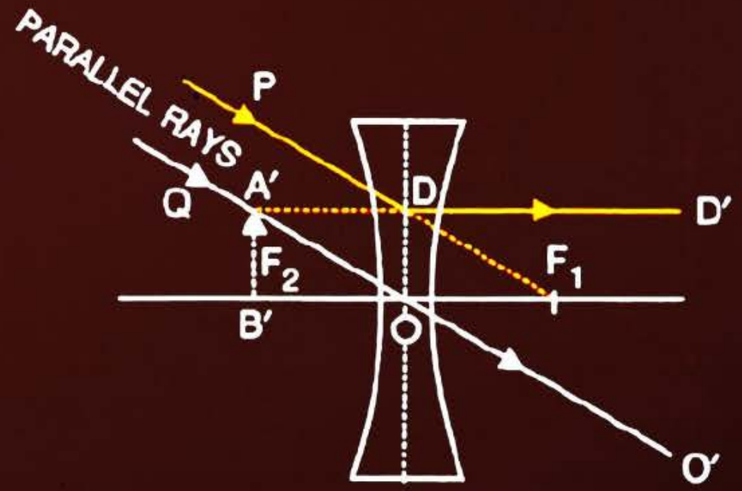
(i) at focus (F),

(ii) virtual and erect, and

(iii) highly diminished (much smaller than the object).

Application: A concave lens is used in Galilean telescope in this manner.





RELATIVE POSITIONS OF THE OBJECT AND IMAGE IN A CONCAVE LENS

	Position of Object	Position of Image	Nature of the Image	Size of the Image	Application
1.	At infinity	At the focus, on the same side of the lens as the object	Virtual and upright	Highly diminished	Galilean telescope
2.	At any position between infinity and optical centre	Between the focus and optical centre, on the same side of the lens as the object	Virtual and upright	Diminished	Myopic eye

DIFFERENCE BETWEEN THE IMAGE FORMED BY A CONVEX AND A CONCAVE LENS

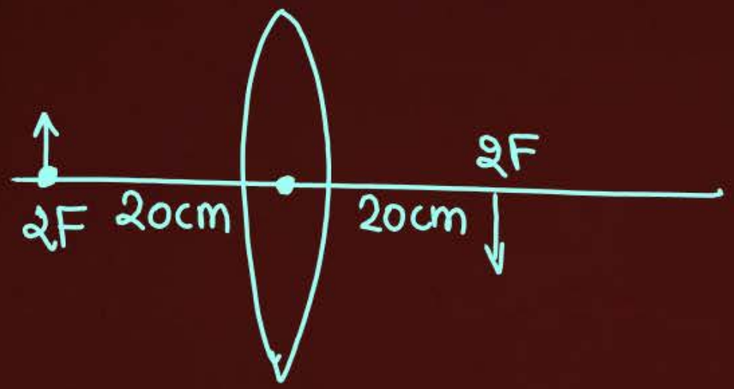
HW

	Image by a Convex Lens		Image by a Concave Lens
1.	The image can be real as well as virtual. It is real if the object lies beyond the focus, while it is virtual if the object lies before the focus.	**1.**	The image is always virtual for all positions of object.
2.	The image can be magnified, of same size as well as diminished. It is magnified if the object lies before $2F$, of same size if the object is at $2F$, and diminished if the object is beyond $2F$.	**2.**	The image is always diminished.
3.	The image can be inverted as well as erect. The image is inverted if the object is beyond focus, and erect if the object is before the focus.	**3.**	The image is always erect.

QUESTIONS

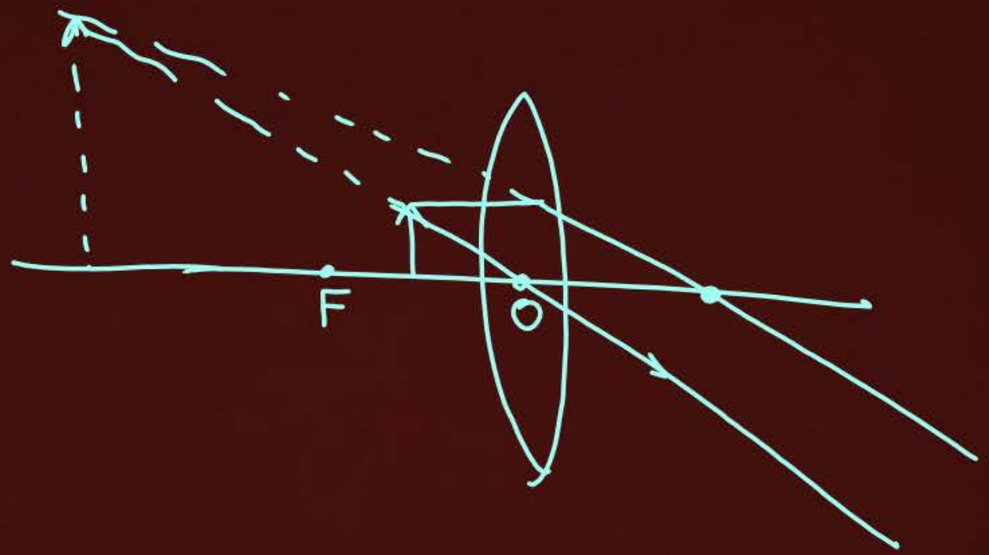
For an object placed at a distance 20 cm in front of a convex lens, the image is at a distance 20 cm behind the lens. The focal length of convex lens is:

1. 20 cm
2. 10 cm
3. 15 cm
4. 40 cm



QUESTIONS

For the object placed between the optical centre and focus of a convex lens, the image is:



1. real and enlarged
2. real and diminished
3. virtual and enlarged
4. virtual and diminished.

QUESTIONS

A concave lens forms the image of an object which is:

1. virtual, inverted, and diminished
2. virtual, upright, and diminished
3. virtual, inverted, and enlarged
4. virtual, upright, and enlarged.

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

Thank You